(b) Using this classification, classify finite subgroups of $SU(2)$ (use the homomorphism $SU(2) \to SO(3)$).

Problem 4.12.9. Find the characters and tensor products of irreducible complex representations of the Heisenberg group from Problem 4.12.2.

Problem 4.12.10. Let G be a finite group and let V be a complex representation of G which is faithful, i.e., the corresponding map $G \to GL(V)$ is injective. Show that any irreducible representation of G occurs inside $S^n V$ (and hence inside $V^{\otimes n}$) for some n.

Hint: Show that there exists a vector $u \in V^*$ whose stabilizer in G is 1. Now define the map $SV \to F(G, \mathbb{C})$ sending a polynomial f on V^* to the function f_u on G given by $f_u(g) = f(gu)$. Show that this map is surjective and use this to deduce the desired result.

Problem 4.12.11. This problem is about an application of representation theory to physics (elasticity theory). We first describe the physical motivation and then state the mathematical problem.

Imagine a material which occupies a certain region U in the physical space $V = \mathbb{R}^3$ (a space with a positive definite inner product). Suppose the material is deformed. This means, we have applied a diffeomorphism (= change of coordinates) $g : U \to U'$. The question in elasticity theory is how much stress in the material this deformation will cause.

For every point $P \in U$, let $A_P : V \to V$ be defined by $A_P = dg(P)$. Here A_P is nondegenerate, so it has a polar decomposition $A_P = D_P O_P$, where O_P is orthogonal and D_P is symmetric. The matrix O_P characterizes the rotation part of A_P (which clearly produces no stress), and D_P is the distortion part, which actually causes stress. If the deformation is small, D_P is close to 1, so $D_P = 1 + d_P$, where d_P is a small symmetric matrix, i.e., an element of $S^2 V$. This matrix is called the **deformation tensor** at P.

Now we define the stress tensor, which characterizes stress. Let v be a small nonzero vector in V, and let σ be a small disk perpendicular to v centered at P of area $||v||$. Let F_v be the force with which the part of the material on the v-side of σ acts on the part on the opposite